Stanisław Szufa

Curriculum Vitae

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Research area. Computational social choice: algorithms and data for collective decisions. I develop methods to compare elections and to test voting rules on real and synthetic data, I built the largest open dataset of participatory budgeting votes, and I now study whether AI-written summaries and LLM agents represent the opinions they aggregate.

Academic Positions

2026 – now **Postdoctoral Researcher**, University of Geneva
Hosts: Roland Bouffanais and Didier Wernli. NLP pipelines for diplomatic and policy texts; computational models of decision-making in multilateral negotiations; LLM-based simulation of democratic deliberation.

2024 – 2026 **CNRS Postdoctoral Researcher**, LAMSADE, Université Paris Dauphine-PSL
Host: Jérôme Lang. Social choice and machine learning; discovering hidden structure in preference data.

2021 – 2024 **Research Assistant**, AGH University of Kraków
ERC project PRAGMA led by Piotr Faliszewski. Maps of elections, participatory budgeting, development of `prefsampling`.

2018 – 2021 **Teaching and Research Assistant**, Jagiellonian University, Kraków
Taught three undergraduate courses; founded Pabulib.

Education

2017 – 2024 **PhD in Computer Science**, AGH University of Kraków
Doctoral studies at Jagiellonian University (2017–2023); thesis defended with distinction at AGH on 13 May 2024. Thesis: *Map of Elections*. Advisor: Piotr Faliszewski.

2015 – 2017 **MSc in Computer Science**, Jagiellonian University

2012 – 2015 **BSc in Computer Science**, Jagiellonian University

Awards, Fellowships, and Grants

2025 **Polish Prime Minister's Award**
For an outstanding doctoral dissertation (*Map of Elections*).

2023 **Pragnesh Jay Modi Best Student Paper Award**, AAMAS-2023
For *A Map of Diverse Synthetic Stable Roommates Instances*, with N. Boehmer and K. Heeger.

2023 **START Fellowship**, Foundation for Polish Science

2018 **Principal Investigator, NCN Preludium grant**, National Science Centre, Poland
Project: *Election Isomorphism Problem*. The project that started maps of elections.

Publications

5 journal articles (*Artificial Intelligence*, *JAIR* ×2, *JCSS*, *Theory and Decision*) and 28 full conference papers (IJCAI ×11, AAMAS ×7, AAAI ×5, NeurIPS ×2, ECAI ×2, Findings of EMNLP ×1). The full list is provided as a separate document, with the same labels as below. Selected papers:

[J4] S. Szufa, N. Boehmer, R. Bredereck, P. Faliszewski, R. Niedermeier, P. Skowron, A. Slinko, N. Talmon. Drawing a Map of Elections. *Artificial Intelligence* 343, 2025.

[C7] S. Szufa, P. Faliszewski, Ł. Janeczko, M. Lackner, A. Slinko, K. Sornat, N. Talmon. How to Sample Approval Elections? *IJCAI-2022*.

[C11] P. Faliszewski, J. Flis, D. Peters, G. Pierczyński, P. Skowron, D. Stolicki, S. Szufa, N. Talmon. Participatory Budgeting: Data, Tools and Analysis. *IJCAI-2023*.

[J1] N. Boehmer, K. Heeger, S. Szufa. A Map of Diverse Synthetic Stable Matching Instances. *JAIR* 79, 2024 (journal version of the AAMAS-2023 Best Student Paper).

Authors in computational social choice are usually listed alphabetically; papers that deviate from alphabetical order list the lead author first.

Software and Datasets

2020 – now **Pabulib**, pabulib.org
Founder. The largest open library of participatory budgeting data: 2,053 elections from 30 cities in 8 countries, 8.6 million votes; cited in more than 100 papers.

prefsampling, Python library
Co-developer, with Simon Rey. Sampling synthetic preference data under a wide range of statistical cultures.

mapof, Python library
Developer. Computing distances between elections and drawing maps of elections.

Teaching and Supervision

2018 – 2021 **Undergraduate courses**, Jagiellonian University
Introduction to Programming (one year), *Logic and Set Theory* and *Linear Algebra* (one semester each); groups of about 20 students.

2023 – 2026 **Tutorials and workshops**
Experiments in Social Choice, tutorial at IJCAI 2023, AAMAS 2023, and AAAI 2024.
Computational Tools for Measuring Diversity, Agreement, and Polarization, hands-on workshop at the Data Science Institute, University of Chicago, 2026.

Co-supervision
More than ten BSc and MSc students.

Service and Leadership

2026 – now **Editorial board member**, Journal of Artificial Intelligence Research (JAIR)

Program committee member, NeurIPS, AAAI, AAMAS, IJCAI

Journal reviewer, JCSS, TCS, SCW

2025 – now **Co-founder**, Nauka dla Demokracji (Science for Democracy)

2021 **President**, Jagiellonian University PhD Students' Association

2020 **Representative**, Polish Association of Doctoral Candidates

Research Visits

2026 **University of Chicago**, Chicago, USA

2025 **Northwestern University**, Evanston, USA

2024 **University of Oxford**, Oxford, UK

2023 **University of Warsaw**, Warsaw, Poland

2023 **National University of Singapore**, Singapore

2021 **TU Berlin**, Berlin, Germany

Invited Talks

2025 **Dagstuhl Seminar 25401**, Societal Impact of Computational Social Choice, Germany

2024 **Participatory Budgeting Workshop**, Society for Social Choice and Welfare

2022 **Computational and Topological Social Choice Workshop**, Society for Social Choice and Welfare

Other Experience

2020 **Optimization Consultant**, PCG Academia
Integer linear programming models for real-world resource-allocation problems for a private-sector client.

Languages

Polish (native), English (fluent), Spanish (intermediate), French (conversational).